

DATABASE MANAGEMENT SYSTEMS

Course	B.Tech.-III-Sem.	L	T	P	C
Course Code	25CSPC32	3	-	-	3

Course Outcomes (COs)

COs	Upon completion of course the students will be able to	SDGs
CO1	design simple databases using database architectures	4,9
CO2	apply relational model, algebra, and calculus for data queries	4,9
CO3	implement SQL queries and optimize database performance	4,9
CO4	analyze schema design using normalization techniques	4,9
CO5	evaluate transaction management and NoSQL databases	4,9

Syllabus

Unit	Title/Topics	Hours
I	Introduction to Database Systems	8
Introduction: Introduction and applications of DBMS, Purpose of database, database architecture and structure - abstraction levels, data independence, database languages, database users and DBA roles. Introduction to Database Design: Database design process, data models, ER diagrams - entities, attributes, relationships, constraints, keys, generalization, specialization, aggregation, conceptual design with the E-R model for large enterprise.		
II	Relational Model, Algebra and Calculus	8
The Relational Model: Relational model concepts: schema, tuples, attributes. Integrity constraints (domain, key, referential). Querying relational data. Logical database design: ER to relational mapping, introduction to views. Creation, altering and dropping (destroying) of tables and views. Relational Algebra and Calculus: Relational algebra operators: Selection, projection, union, intersection, difference, joins. Relational calculus: tuple and domain relational calculus.		
III	SQL	4+4=8
Part-A: Basics of SQL, DDL, DML, DCL, structure – creation, alteration, defining constraints – Primary key, foreign key, unique, not null, check, in operator, Functions - aggregate functions, built-in functions – numeric, date, string functions, set operations.		
Part-B: Sub-queries, correlated sub-queries, Use of group by, having, order by, join and its types, exist, any, all, view and its types. Transaction control commands – commit, rollback, save point, cursors, stored procedures, Triggers. Indexing (B+ tree concept, index usage). Query optimization basics (execution plans, performance tuning).		
IV	Schema Refinement and Normal Forms	8
Schema Refinement and Normal Forms: Introduction to schema refinement, functional dependencies, reasoning about FDs. Normalization, normal forms: 1NF, 2NF, 3NF, BCNF. 4NF & 5NF – overview only. Decomposition: Lossless join, Dependency preservation.		
V	Transactions Management and NoSQL Databases	8
Transactions Management: Transaction concept and ACID properties, transaction state, Concurrent execution of transactions, Serializability and testing for Serializability, recoverability. Concurrency Control and Recovery System: Concurrency control, lock based protocols, time-stamp based protocols, validation based protocols, deadlocks (basic idea), crash recovery and backup system. Introduction to NoSQL databases: MongoDB, key-value, document, graph models. Use cases of NoSQL vs RDBMS (comparison).		
Textbooks		
1. Database Management Systems by Raghu Ramakrishnan and Johannes Gehrke, TMH, 2003. 2. Database System Concepts by Abraham Silberschatz, Henry F. Korth and S. Sudarshan, TMH, 2005.		
References		
1. Fundamentals of Database Systems, Elmasri Navate, Pearson Education, India. 2. Oracle SQL, P. K. Das Guptha and P Radha Krishna PHI.		

CO-PO-PSO Mapping (3-Strong; 2-Medium; 1-Weak Correlation)

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	3	2	3	2	2	0	0	0	0	0	1	3	2
CO2	3	3	2	2	2	0	0	0	0	0	1	3	2
CO3	3	3	3	3	3	0	0	0	0	1	1	3	3
CO4	3	3	3	3	2	0	0	0	0	0	1	3	2
CO5	3	3	2	3	3	0	1	0	0	1	2	3	3

CO-PO-PSO Justification

COs	CO-PO Justification	CO-PSO Justification
CO1	Designing simple databases using architectures strongly supports engineering knowledge (PO1) and solution development (PO3). ER modelling and database design involve problem analysis (PO2), investigation of requirements (PO4), and use of database design tools (PO5).	Strongly supports database design as part of software system development (PSO1) and application of data management technologies (PSO2).
CO2	Applying relational model, algebra, and calculus strongly supports engineering knowledge (PO1), problem analysis (PO2), and formal query formulation (PO3). Relational operations involve investigation of data retrieval logic (PO4).	Strongly supports relational data modelling in software systems (PSO1) and application of database query techniques in real-world systems (PSO2).
CO3	Implementing SQL queries and optimizing performance strongly supports problem analysis (PO2), solution development (PO3), investigation of query execution (PO4), and modern tool usage (PO5). Performance tuning also supports project efficiency awareness (PO10).	Strongly supports implementation of database-driven software systems (PSO1) and application of modern DBMS technologies for efficient real-world solutions (PSO2).
CO4	Analyzing schema design using normalization strongly supports problem analysis (PO2), solution development (PO3), and investigation of dependency and redundancy issues (PO4). It strengthens engineering knowledge of database design principles (PO1).	Strongly supports development of reliable and well-structured database systems (PSO1) and application of systematic design techniques in software solutions (PSO2).
CO5	Evaluating transaction management and NoSQL databases supports engineering knowledge (PO1), problem analysis (PO2), and investigation of concurrency, recovery, and distributed data models (PO4). NoSQL tools support modern tool usage (PO5), ethics in data handling (PO7), project reliability (PO10), and lifelong learning (PO11).	Strongly supports development of reliable database-backed software systems (PSO1) and application of advanced database technologies such as MongoDB and NoSQL models (PSO2).

DATABASE MANAGEMENT SYSTEMS LAB

Course	B.Tech.-III-Sem.	L	T	P	C
--------	------------------	---	---	---	---

Course Code	25CSPC35	-	-	2	1
-------------	----------	---	---	---	---

Course Outcomes (COs)

COs	Upon completion of course the students will be able to	SDGs
CO1	create relational databases using SQL and constraints	4,9
CO2	perform data operations and queries using SQL	4,9
CO3	implement views, triggers, and procedures	4,9
CO4	apply normalization and build database applications	4,9
CO5	use MongoDB and integrate SQL with NoSQL systems	4,9

List of Experiments

Note: Take any database application and conduct experiments to get expertise on various case studies

Week	Title/Experiment
1	Creation of Tables using SQL: Data types, DDL commands (CREATE, ALTER, DROP) with primary and foreign keys.
2	DML operations: INSERT, SELECT, UPDATE, DELETE on tables.
3	SQL queries using operators: ANY, ALL, IN, EXISTS, NOT EXISTS, UNION, INTERSECT, EXCEPT, constraints.
4	Sub Queries: Nested and Correlated; Joins: Inner and Outer.
5	Aggregate functions (COUNT, SUM, AVG, MAX, MIN), GROUP BY, HAVING; creation and management of views.
6	Triggers: Creation and execution of INSERT, UPDATE, DELETE triggers.
7	Stored procedures: Create, execute, and modify stored procedures.
8	Application: Develop a mini database application using SQL.
9	Normalization: Convert given tables into 1NF, 2NF, and 3NF.
10	NoSQL database (MongoDB): Perform CRUD operations on collections.
11	NoSQL Database: Implement arrays, functions (count, sort, aggregate) using MongoDB.
12	Integration Task: Build a simple application integrating SQL and MongoDB operations.
References	
1. Database Management Systems Lab Manual, Department of CSE, CMRIT, Hyd.	
Micro-Projects: Student should submit a report on <i>design, implement, test and debug</i> from one of the following/any other micro-project(s) approved by the faculty before the deadline.	
1. Student Information Management System using SQL and Relational Database. 2. Online Book Store Database with Order and Payment Management. 3. Hospital Management Database System with Patient Records and Billing. 4. Inventory Control System with Supplier and Stock Management. 5. Employee Payroll Database with Triggers and Stored Procedures. 6. Railway Reservation System with Seat Allocation and Ticket Booking. 7. Online Food Ordering System with Menu, Orders, and Delivery Tracking. 8. Banking Database System with Transactions and Account Management. 9. Social Media Data Management using NoSQL (MongoDB). 10. E-Commerce Product Catalog using MongoDB with Aggregation Queries.	

Test Cases

Week	Test Cases for Experiments			
1	Table Creation using SQL			
	Type	Input	Expected Output	Explanation
	Success	Create Student table	Table created	Valid table creation
	Success	Create FK relationship	Tables linked	Foreign key validation
	Success	Add new column	Column added	ALTER operation
	Failure/Edge	Duplicate primary key	Error displayed	Constraint validation
	Failure/Edge	Missing referenced table	Error displayed	FK dependency
2	DML Operations			
	Type	Input	Expected Output	Explanation
	Success	Insert student record	Record inserted	Create operation
	Success	Select student records	Records displayed	Read operation
	Success	Update marks	Record updated	Update operation
	Failure/Edge	NULL in NOT NULL field	Error displayed	Constraint validation
	Failure/Edge	Delete unmatched row	0 rows affected	Safe delete
3	SQL Operators and Set Operations			
	Type	Input	Expected Output	Explanation
	Success	Query using IN	Matching rows displayed	IN operator
	Success	Query using EXISTS	Valid rows displayed	EXISTS operator
	Success	UNION query	Combined rows shown	Set operation
	Failure/Edge	UNION mismatch columns	Error displayed	Compatibility check
	Failure/Edge	Empty subquery	Logical result shown	Edge condition
4	Subqueries and Joins			
	Type	Input	Expected Output	Explanation
	Success	Nested subquery	Correct rows displayed	Subquery execution
	Success	INNER JOIN query	Matching rows shown	Join validation
	Success	LEFT JOIN query	NULL rows included	Outer join
	Failure/Edge	Join without ON	Unexpected rows displayed	Logical mistake
	Failure/Edge	Incorrect subquery condition	Incorrect output shown	Weak logic
5	Aggregate Functions and Views			
	Type	Input	Expected Output	Explanation
	Success	COUNT with GROUP BY	Group counts displayed	Aggregation
	Success	HAVING clause query	Filtered groups shown	HAVING validation
	Success	Create VIEW	View created	View management
	Failure/Edge	Invalid GROUP BY query	Error displayed	Aggregation rule
	Failure/Edge	Drop base table	View invalid	Dependency issue
6	Triggers			
	Type	Input	Expected Output	Explanation
	Success	INSERT trigger	Log updated	Insert trigger
	Success	UPDATE trigger	Audit row created	Update trigger
	Success	DELETE trigger	Backup stored	Delete trigger
	Failure/Edge	Invalid trigger column	Error displayed	Schema validation
	Failure/Edge	Recursive trigger update	Recursion error displayed	Logical issue
7	Stored Procedures			
	Type	Input	Expected Output	Explanation
	Success	Procedure insert	Record inserted	Procedure execution
	Success	Average calculation procedure	Average displayed	Output procedure
	Success	Update procedure	Record updated	Procedure modification
	Failure/Edge	Missing parameter	Error displayed	Parameter validation
	Failure/Edge	Duplicate key insertion	Constraint error	Data integrity

8	Mini Database Application using SQL			
	Type	Input	Expected Output	Explanation
	Success	Add student data	Record saved	Application insert
	Success	Search student	Record displayed	Application query
	Success	Update student details	Record updated	Application update
	Failure/Edge	Invalid RollNo	No record found	Input validation
	Failure/Edge	SQL injection input	Query blocked	Security validation
9	Normalization			
	Type	Input	Expected Output	Explanation
	Success	Repeating attributes	Converted to 1NF	Removes redundancy
	Success	Partial dependency	Converted to 2NF	Dependency removal
	Success	Transitive dependency	Converted to 3NF	Normalization
	Failure/Edge	Already normalized table	No decomposition	Avoids redundancy
	Failure/Edge	Lossy decomposition	Rejected decomposition	Lossless design
10	MongoDB CRUD Operations			
	Type	Input	Expected Output	Explanation
	Success	insertOne() document	Document inserted	Create operation
	Success	find() query	Documents displayed	Read operation
	Success	updateOne() query	Document updated	Update operation
	Failure/Edge	deleteOne() unmatched	0 documents deleted	Safe delete
	Failure/Edge	Duplicate _id	Error displayed	Constraint handling
11	MongoDB Arrays, Count, Sort, Aggregate			
	Type	Input	Expected Output	Explanation
	Success	Array document	Array stored	Array handling
	Success	countDocuments() query	Count displayed	Count operation
	Success	sort() query	Sorted output shown	Sorting validation
	Failure/Edge	Empty collection aggregate	Empty result shown	Boundary case
	Failure/Edge	Missing sort field	Output generated	Schema-less handling
12	Sorting			
	Type	Input	Expected Output	Explanation
	Success	Store student data in SQL	Record saved	Relational storage
	Success	Store activity logs in MongoDB	Document inserted	NoSQL storage
	Success	Fetch combined student profile	Combined output displayed	Database integration
	Failure/Edge	Missing MongoDB record	Partial profile shown	Missing data handling
	Failure/Edge	Invalid student ID	Error displayed	Invalid retrieval

CO-PO-PSO Mapping (3-Strong; 2-Medium; 1-Weak Correlation)

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	3	2	3	1	3	0	0	1	1	0	1	3	2
CO2	3	3	3	2	3	0	0	1	1	0	1	3	2
CO3	3	3	3	2	3	0	0	1	1	0	1	3	2
CO4	3	3	3	3	2	0	0	1	1	0	1	3	2
CO5	3	3	3	3	3	0	0	1	1	0	1	3	3

CO-PO-PSO Justification

COs	CO-PO Justification	CO-PSO Justification
CO1	Creating relational databases using SQL and constraints supports engineering knowledge (PO1), problem analysis (PO2), and strong solution development (PO3). Use of SQL tools strongly supports modern tool usage (PO5).	Strongly supports database design and development in software systems (PSO1) and application of relational database technologies (PSO2).
CO2	Performing data operations and queries strongly supports problem analysis (PO2), solution development (PO3), and investigation of query results (PO4). SQL execution strongly supports tool usage (PO5).	Strongly supports implementation of database operations in software systems (PSO1) and application of query processing techniques (PSO2).
CO3	Implementing views, triggers, and procedures supports engineering knowledge (PO1), problem analysis (PO2), and solution development (PO3). These involve investigation of database behavior (PO4) and use of advanced DBMS tools (PO5).	Strongly supports development of dynamic and automated database systems (PSO1) and application of advanced database features (PSO2).
CO4	Applying normalization and building database applications strongly supports problem analysis (PO2), solution development (PO3), and investigation of data redundancy and integrity (PO4).	Strongly supports development of structured and efficient database systems (PSO1) and application of design principles in real-world systems (PSO2).
CO5	Using MongoDB and integrating SQL with NoSQL systems strongly supports problem analysis (PO2), solution development (PO3), and investigation of modern data systems (PO4). Strong use of tools like MongoDB supports PO5.	Strongly supports development of hybrid database-driven applications (PSO1) and strong application of cutting-edge NoSQL technologies (PSO2).